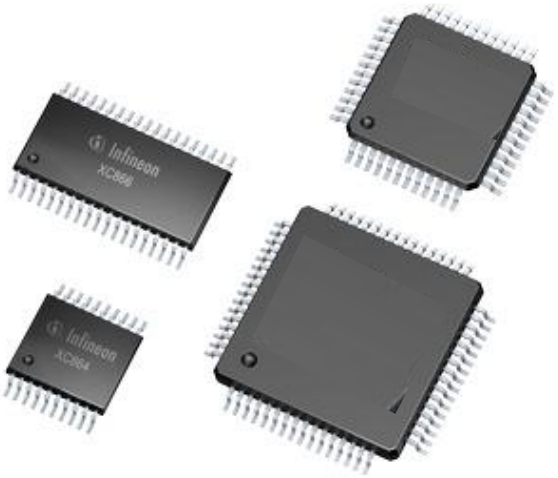
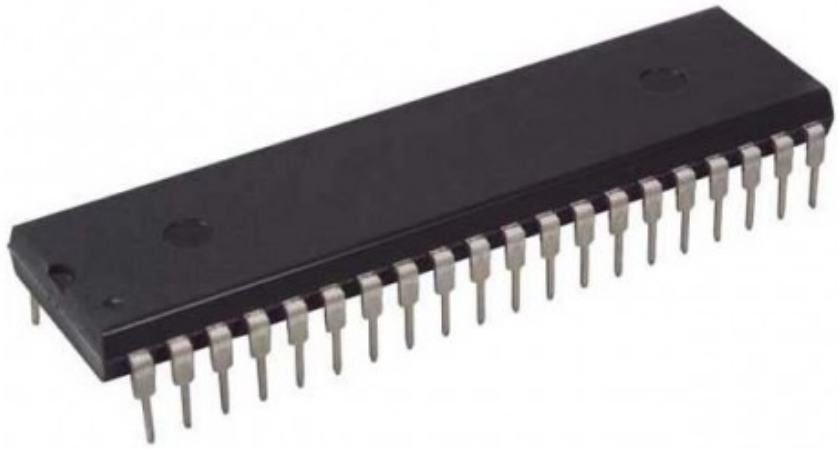
8051 Microcontroller





Assembly language instruction consist of 4 fields

Label:	**mnemonic**	**operand**	**;comment**
HERE:	MOV	A,#6AH	;mov operation

Arithmetic Instruction set

Mnemonic	Description	Byte	Cycle
Arithmetic operations			
ADD A,Rn	Add register to accumulator	1	1
ADD A,direct	Add direct byte to accumulator	2	1
ADD A,@Ri	Add indirect RAM to accumulator	1	1
ADD A,#data	Add immediate data to accumulator	2	1
ADDC A,Rn	Add register to accumulator with carry flag	1	1
ADDC A,direct	Add direct byte to A with carry flag	2	1
ADDC A,@Ri	Add indirect RAM to A with carry flag	1	1
ADDC A,#data	Add immediate data to A with carry flag	2	1

Arithmetic Instruction set

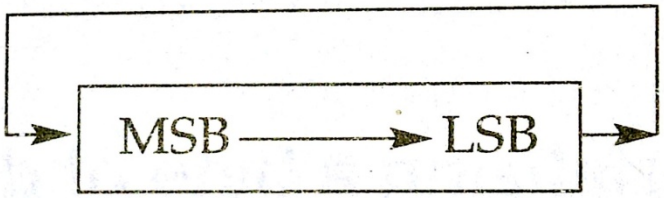
Mnemonic	Description	Byte	Cycle
Arithmetic operations			
SUBB A,Rn	Subtract register to accumulator with borrow	1	1
SUBB A,direct	Subtract direct byte to A with carry borrow	2	1
SUBB A,@Ri	Subtract indirect RAM to A with carry borrow	1	1
SUBB A,#data	Subtract immediate data to A with carry borrow	2	1
INC A	Increment accumulator	1	1
INC Rn	Increment register	1	1
INC direct	Increment direct byte	2	1
INC @Ri	Increment indirect RAM	1	1
INC DPTR	Increment data pointer	1	2

Arithmetic Instruction set

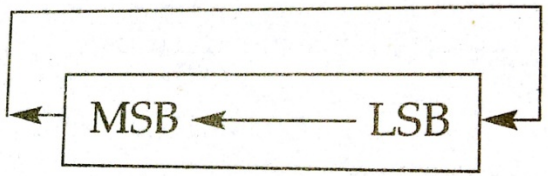
Mnemonic		Description	Byte	Cycle
Arithmetic operations				
DEC	A	Decrement accumulator	1	1
DEC	Rn	Decrement register	1	1
DEC	direct	Decrement direct byte	2	1
DEC	@Ri	Decrement indirect RAM	1	1
MUL	AB	Multiply A and B -> [B hi]:[A lo]	1	4
DIV	AB	Divide A by B -> A=result, B=remainder	1	4
DA	A	Decimal adjust accumulator	1	1

Rotate Instruction set

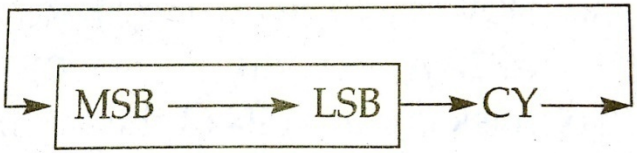
Mnemonic	Description	Byte	Cycle
RL A	Rotate accumulator left	1	1
RLC A	Rotate accumulator left through carry	1	1
RR A	Rotate accumulator right	1	1
RRC A	Rotate accumulator right through carry	1	1
SWAP A	Swap nibbles within the accumulator	1	1



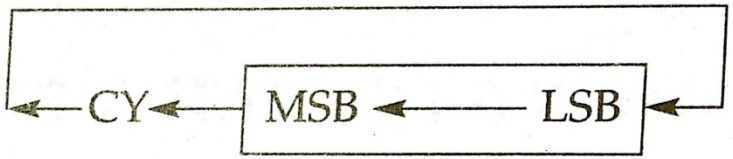
RR



RL



RRC



RLC

Logical Instruction set

Mnemonic		Description	Byte	Cycle
ANL	A,Rn	AND register to accumulator	1	1
ANL	A,direct	AND direct byte to accumulator	2	1
ANL	A,@Ri	AND indirect RAM to accumulator	1	1
ANL	A,#data	AND immediate data to accumulator	2	1
ANL	direct,A	AND accumulator to direct byte	2	1
ANL	direct,#data	AND immediate data to direct byte	3	2

Mnemonic		Description	Byte	Cycle
ORL	A,Rn	OR register to accumulator	1	1
ORL	A,direct	OR direct byte to accumulator	2	1
ORL	A,@Ri	OR indirect RAM to accumulator	1	1
ORL	A,#data	OR immediate data to accumulator	2	1
ORL	direct,A	OR accumulator to direct byte	2	1
ORL	direct,#data	OR immediate data to direct byte	3	2
XRL	A,Rn	Exclusive OR register to accumulator	1	1
XRL	A,direct	Exclusive OR direct byte to accumulator	2	1
XRL	A,@Ri	Exclusive OR indirect RAM to accumulator	1	1
XRL	A,#data	Exclusive OR immediate data to accumulator	2	1
XRL	direct,A	Exclusive OR accumulator to direct byte	2	1
XRL	direct,#data	Exclusive OR immediate data to direct byte	3	2

Bit manipulation Instruction set

Mnemonic		Description	Byte	Cycle
CLR	C	Clear carry flag	1	1
CLR	bit	Clear direct bit	2	1
SETB	C	Set carry flag	1	1
SETB	bit	Set direct bit	2	1
CPL	C	Complement carry flag	1	1
CPL	bit	Complement direct bit	2	1
ANL	C,bit	AND direct bit to carry flag	2	2
ANL	C,/bit	AND complement of direct bit to carry	2	2
ORL	C,bit	OR direct bit to carry flag	2	2
ORL	C,/bit	OR complement of direct bit to carry	2	2
MOV	C,bit	Move direct bit to carry flag	2	1
MOV	bit,C	Move carry flag to direct bit	2	2

8051 microcontroller

0400h	MOV TMOD, #01H
0402h	**again1**: MOV TL0, #00H
0404h	MOV TH0, #0EEH
0406h	MOV P1, #0FFH
0408h	ACALL **DELAY**
040Ah	MOV P1, #00H
040Ch	ACALL **DELAY**
040Eh	SJMP **again1**

PC → 040Ah

Main code

PC → DELAY: SETB TR0

again: JNB TF0 again

CLR TR0

CLR TF0

PC → RET

Sub-routine

PC=040Ah

Address RAM	Stack memory
0007h	0A
0008h	04
0009h	00

Address RAM	Stack memory
0007h	0A
0008h	04
0009h	00

Program and Machine control Instruction set

Mnemonic	Description	Byte	Cycle
ACALL addr11	Absolute subroutine call	2	2
LCALL addr16	Long subroutine call	3	2
RET	Return from subroutine	1	2
RETI	Return from interrupt	1	2
AJMP addr11	Absolute jump	2	2
LJMP addr16	Long jump	3	2
SJMP rel	Short jump (relative address)	2	2

LCALL

Long call

- 3 byte instruction
- Address: 16 bit

- LCALL can be used to call subroutines located anywhere within 64 Kbyte address space of 8051

ACALL

Absolute call

- 2 byte instruction
- Address: 11 bit

- ACALL can be used to call subroutines located within 2 Kbyte address space of 8051

LJMP

Long jump

- 3 byte instruction
- Address: 16 bit
- LJMP can be used to jump anywhere within 64 Kbyte address space of 8051

SJMP

Short jump

- 2 byte instruction
- 1st byte: opcode
- 2nd byte: relative address
- Address: 8 bit therefore relative address range= 00 to FFh
- Is divided into forward and backward jump, that is within -128 to 127 bytes of memory relative to address of current PC.

AJMP

Long jump

- 2 byte instruction
- Address: 11 bit
- AJMP can be used to jump anywhere within 2 Kbyte address space of 8051

PC	OPCODE		CODE
0002H	7455		MOV A,#55H
0004H	6003		JZ NEXT
0006H	08		INC R0
0007H	04		INC A
0008H	04		INC A
0009H	2477	NEXT:	ADD A, #77H

Relative address= target jump address- PC

PC	OPCODE		CODE
0002H	7455		MOV A,#55H
0004H	04	NEXT:	INC A
0005H	08		INC R0
0006H	04		INC A
0007H	60FB		JZ NEXT
0009H	2477		ADD A, #77H

Program and Machine control Instruction set

Mnemonic		Description	Byte	Cycle
JMP	@A+DPTR	Jump indirect relative to the DPTR	1	2
JZ	rel	Jump if accumulator is zero	2	2
JNZ	rel	Jump if accumulator is not zero	2	2
JC	rel	Jump if carry flag is set	2	2
JNC	rel	Jump if carry flag is not set	2	2
JB	bit,rel	Jump if bit is set	3	2
JNB	bit,rel	Jump if bit is not set	3	2
JBC	bit,rel	Jump if direct bit is set and clear bit	3	2
CJNE	A,direct,rel	Compare direct byte to A and jump if not equal	3	2

Program and Machine control Instruction set

Mnemonic	Description	Byte	Cycle
Mnemonic	Description	Byte	Cycle
CJNE A,#data,rel	Compare immediate to A and jump if not equal	3	2
CJNE Rn,#data,rel	Compare immed. to reg. and jump if not equal	3	2
CJNE @Rn,#data,rel	Compare immed. to ind. and jump if not equal	3	2
DJNZ Rn,rel	Decrement register and jump in not zero	2	2
DJNZ direct,rel	Decrement direct byte and jump in not zero	3	2
NOP	No operation	1	1

Data transfer Instruction set

Mnemonic		Description	Byte	Cycle
MOV	A,Rn	Move register to accumulator	1	1
MOV	A,direct*)	Move direct byte to accumulator	2	1
MOV	A,@Ri	Move indirect RAM to accumulator	1	1
MOV	A,#data	Move immediate data to accumulator	2	1
MOV	Rn,A	Move accumulator to register	1	1
MOV	Rn,direct	Move direct byte to register	2	2
MOV	Rn,#data	Move immediate data to register	2	1
MOV	direct,A	Move accumulator to direct byte	2	1
MOV	direct,Rn	Move register to direct byte	2	2
MOV	direct,direct	Move direct byte to direct byte	3	2
MOV	direct,@Ri	Move indirect RAM to direct byte	2	2
MOV	direct,#data	Move immediate data to direct byte	3	2
MOV	@Ri,A	Move accumulator to indirect RAM	1	1
MOV	@Ri,direct	Move direct byte to indirect RAM	2	2
MOV	@Ri,#data	Move immediate data to indirect RAM	2	1
MOV	DPTR,#data16	Load data pointer with a 16-bit constant	3	2

Data transfer Instruction set

Mnemonic	Description	Byte	Cycle
MOVC A,@A+DPTR	Move code byte relative to DPTR to accumulator	1	2
MOVC A,@A+PC	Move code byte relative to PC to accumulator	1	2
MOVX A,@Ri	Move external RAM (8-bit addr.) to A	1	2
MOVX A,@DPTR	Move external RAM (16-bit addr.) to A	1	2
MOVX @Ri,A	Move A to external RAM (8-bit addr.)	1	2
MOVX @DPTR,A	Move A to external RAM (16-bit addr.)	1	2
PUSH direct	Push direct byte onto stack	2	2
POP direct	Pop direct byte from stack	2	2
XCH A,Rn	Exchange register to accumulator	1	1
XCH A,direct	Exchange direct byte to accumulator	2	1
XCH A,@Ri	Exchange indirect RAM to accumulator	1	1
XCHD A,@Ri	Exchange low-order nibble indir. RAM with A	1	1